R6 Friction A level only

Name:

Class:

Date:

Time: **447 minutes**

Marks: **371 marks**

Comments:

Q1.

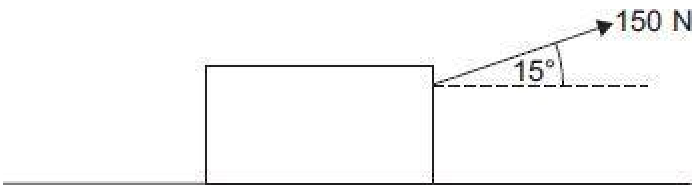
In this question use g = 9.8 m s^{-2}

A boy attempts to move a wooden crate of mass 20 kg along horizontal ground. The coefficient of friction between the crate and the ground is 0.85

(a) The boy applies a horizontal force of 150 N. Show that the crate remains stationary.

(3)

(b) Instead, the boy uses a handle to pull the crate forward. He exerts a force of 150 N, at an angle of 15° above the horizontal, as shown in the diagram.



Determine whether the crate remains stationary.

Fully justify your answer.

(5)

(Total 8 marks)

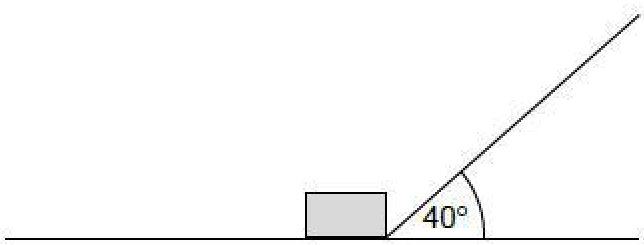
Q2.

In this question use g = 9.8 m s^{-2}, giving your final answers to an appropriate degree of accuracy.

The diagram shows a box, of mass 8.0 kg, being pulled by a string so that the box moves at a constant speed along a rough horizontal wooden board.

The string is at an angle of 40° to the horizontal.

The tension in the string is 50 newtons.



The coefficient of friction between the box and the board is μ

Model the box as a particle.

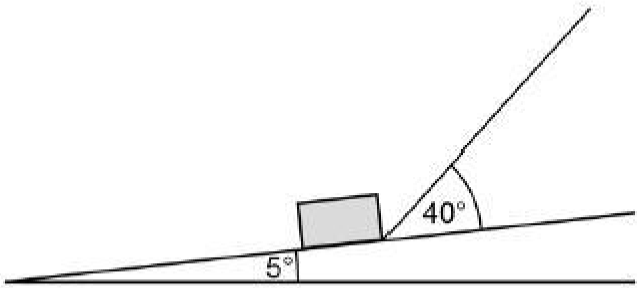
(a) Show that μ = 0.83

(4)

(b) One end of the board is lifted up so that the board is now inclined at an angle of 5° to the horizontal.

The box is pulled up the inclined board.
The string remains at an angle of 40° to the board.

The tension in the string is increased so that the box accelerates up the board at 3 m s^{-2}



(i) Draw a diagram to show the forces acting on the box as it moves.

(1)

(ii) Find the tension in the string as the box accelerates up the slope at 3 m s^{-2}.

(7)

(Total 12 marks)

Q3.

A box, of mass 3 kg, is placed on a rough slope inclined at an angle of 40° to the horizontal. It is released from rest and slides down the slope.

(a) Draw a diagram to show the forces acting on the box.

(1)

(b) Find the magnitude of the normal reaction force acting on the box.

(2)

(c) The coefficient of friction between the box and the slope is 0.2. Find the magnitude of the friction force acting on the box.

(2)

(d) Find the acceleration of the box.

(3)

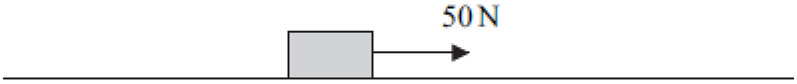
(e) State an assumption that you have made about the forces acting on the box.

(1)

(Total 9 marks)

Q4.

A block, of mass 4 kg, is made to move in a straight line on a rough horizontal surface by a horizontal force of 50 newtons, as shown in the diagram.



Assume that there is no air resistance acting on the block.

(a) Draw a diagram to show all the forces acting on the block.

(1)

(b) Find the magnitude of the normal reaction force acting on the block.

(1)

(c) The acceleration of the block is 3 m s^{-2}. Find the magnitude of the friction force acting on the block.

(3)

(d) Find the coefficient of friction between the block and the surface.

(2)

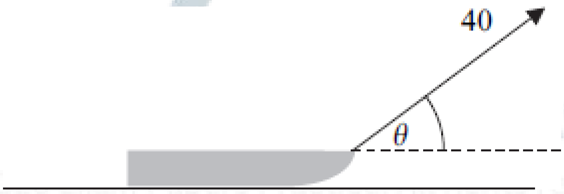
(e) Explain how and why your answer to part (d) would change if you assumed that air resistance did act on the block.

(2)

(Total 9 marks)

Q5.

A child pulls a sledge, of mass 8 kg, along a rough horizontal surface, using a light rope. The coefficient of friction between the sledge and the surface is 0.3. The tension in the rope is 40 newtons. The rope is kept at an angle of θ to the horizontal, as shown in the diagram.



Model the sledge as a particle.

(a) Draw a diagram to show all the forces acting on the sledge.

(1)

(b) Find the magnitude of the normal reaction force acting on the sledge, in terms of θ.

(3)

(c) The acceleration of the sledge is

$$p + q\cos\theta + r\sin\theta$$

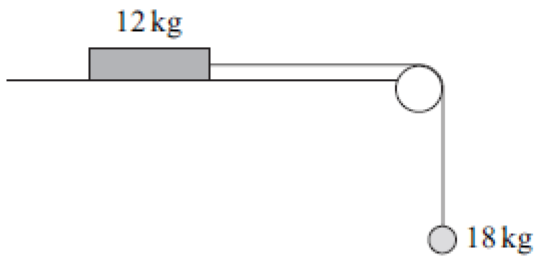
Find p, q and r.

(5)

(Total 9 marks)

Q6.

A block, of mass 12 kg, lies on a horizontal surface. The block is attached to a particle, of mass 18 kg, by a light inextensible string which passes over a smooth fixed peg. Initially, the block is held at rest so that the string supports the particle, as shown in the diagram.



The block is then released.

(a) Assuming that the surface is smooth, use two equations of motion to find the magnitude of the acceleration of the block and particle.

(4)

(b) In reality, the surface is rough and the acceleration of the block is 3 m s^{-2}.

(i) Find the tension in the string.

(3)

(ii) Calculate the magnitude of the normal reaction force acting on the block.

(1)

(iii) Find the coefficient of friction between the block and the surface.

(5)

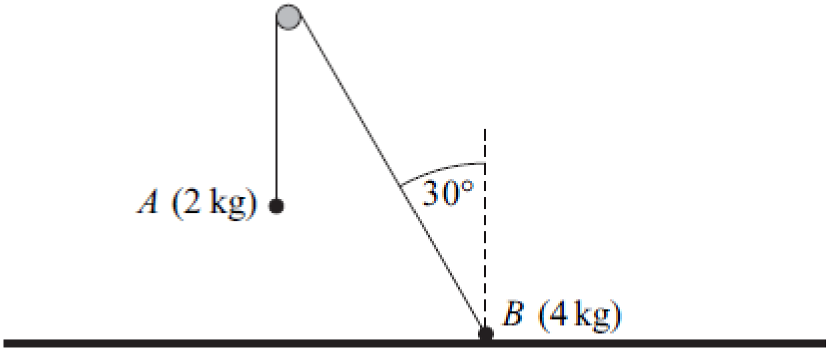
(c) State two modelling assumptions, other than those given, that you have made in answering this question.

(2)

(Total 15 marks)

Q7.

Two particles, A and B, are connected by a light inextensible string which passes over a smooth peg. Particle A has mass 2 kg and particle B has mass 4 kg. Particle A hangs freely with the string vertical. Particle B is at rest in equilibrium on a rough horizontal surface with the string at an angle of 30° to the vertical. The particles, peg and string are shown in the diagram.



(a) By considering particle A, find the tension in the string.

(2)

(b) Draw a diagram to show the forces acting on particle B.

(2)

(c) Show that the magnitude of the normal reaction force acting on particle B is 22.2 newtons, correct to three significant figures.

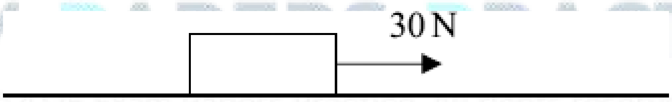
(3)

(d) Find the least possible value of the coefficient of friction between particle B and the surface.

(4)
(Total 11 marks)

Q8.

A wooden block, of mass 4 kg, is placed on a rough horizontal surface. The coefficient of friction between the block and the surface is 0.3 . A horizontal force, of magnitude 30 newtons, acts on the block and causes it to accelerate.



(a) Draw a diagram to show all the forces acting on the block.

(1)

(b) Calculate the magnitude of the normal reaction force acting on the block.

(1)

(c) Find the magnitude of the friction force acting on the block.

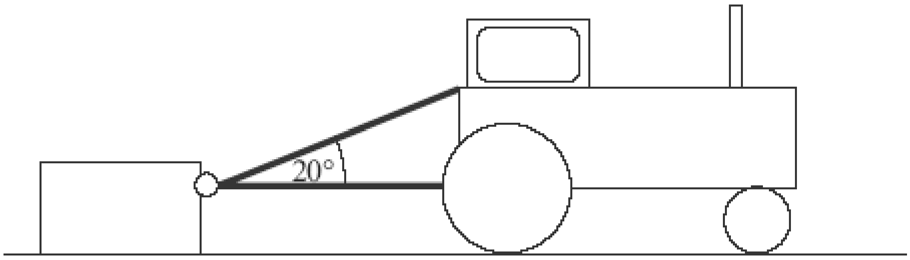
(2)

(d) Find the acceleration of the block.

(3)
(Total 7 marks)

Q9.

A crate, of mass 200 kg, is initially at rest on a rough horizontal surface. A smooth ring is attached to the crate. A light inextensible rope is passed through the ring, and each end of the rope is attached to a tractor. The lower part of the rope is horizontal and the upper part is at an angle of 20° to the horizontal, as shown in the diagram.



When the tractor moves forward, the crate accelerates at 0.3 m s^{-2}. The coefficient of friction between the crate and the surface is 0.4.

Assume that the tension, T newtons, is the same in both parts of the rope.

(a) Draw and label a diagram to show the forces acting on the crate.

(2)

(b) Express the normal reaction between the surface and the crate in terms of T.

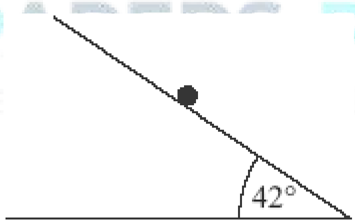
(3)

(c) Find T.

(5)
(Total 10 marks)

Q10.

A particle, of mass m kg, is at rest on a rough plane which is inclined at an angle of 42° to the horizontal, as shown in the diagram.



The friction force acting on the particle has magnitude 30 newtons.

(a) Draw and label a diagram to show all the forces acting on the particle.

(1)

(b) Find m.

(3)

(c) Find the magnitude of the normal reaction force acting on the particle.

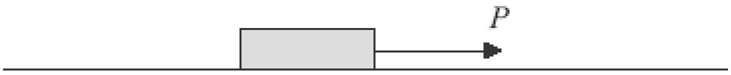
(2)

(d) Given that the particle is on the point of sliding down the plane, find the coefficient of friction between the particle and the plane.

(3)
(Total 9 marks)

Q11.

A block, of mass 10 kg, is at rest on a rough horizontal surface, when a horizontal force, of magnitude P newtons, is applied to the block, as shown in the diagram.



The coefficient of friction between the block and the surface is 0.5.

(a) Draw and label a diagram to show all the forces acting on the block.

(1)

(b) (i) Calculate the magnitude of the normal reaction force acting on the block.

(1)

(ii) Find the maximum possible magnitude of the friction force between the block and the surface.

(1)

(iii) Given that P = 30, state the magnitude of the friction force acting on the block.

(1)

(c) Given that P = 80, find the acceleration of the block.

(3)
(Total 7 marks)

Q12.

A sledge of mass 8 kg is at rest on a rough horizontal surface. A child tries to move the sledge by pushing it with a pole, as shown in the diagram, but the sledge **does not move**. The pole is at an angle of 30° to the horizontal and exerts a force of 40 newtons on the sledge.

Model the sledge as a particle.

(a) Draw a diagram to show the four forces acting on the sledge.

(1)

(b) Show that the normal reaction force between the sledge and the surface has magnitude 98.4 N.

(3)

(c) Find the magnitude of the friction force that acts on the sledge.

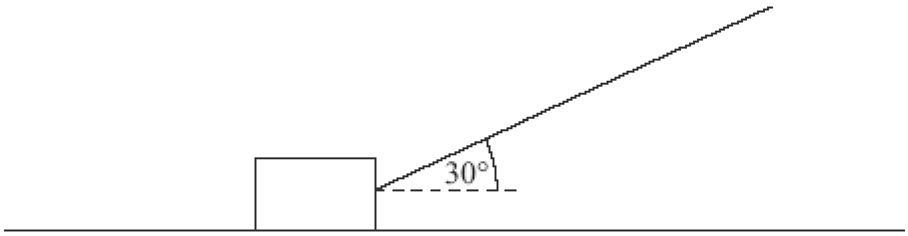
(2)

(d) Find the least possible value of the coefficient of friction between the sledge and the surface.

(3)
(Total 9 marks)

Q13.

The diagram shows a block, of mass 20 kg, being pulled along a rough horizontal surface by a rope inclined at an angle of 30° to the horizontal.



The coefficient of friction between the block and the surface is μ. Model the block as a particle which slides on the surface.

(a) If the tension in the rope is 60 newtons, the block moves at a constant speed.

(i) Show that the magnitude of the normal reaction force acting on the block is 166 N.

(3)

(ii) Find μ.

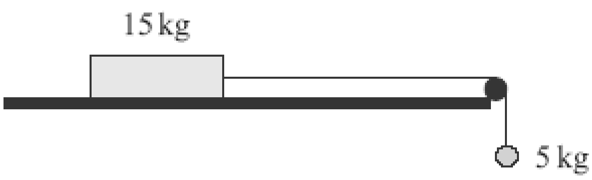
(4)

(b) If the rope remains at the same angle and the block accelerates at 0.8 m s^{-2}, find the tension in the rope.

(5)
(Total 12 marks)

Q14.

A block, of mass 15 kg, is placed on a rough horizontal surface. It is attached, by a light inextensible string that passes over a smooth peg, to a particle of mass 5 kg, which hangs freely, as shown in the diagram.



The coefficient of friction between the block and the surface is 0.2. The block and particle are released from rest and move with the string taut.

(a) Find the magnitude of the friction force acting on the block.

(3)

(b) Show that the acceleration of the system is 0.98 m s^{-2}.

(5)

(c) The block is travelling at 1.4 m s^{-1} when it reaches the peg.

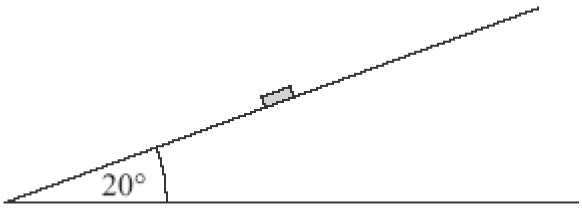
Find the distance that the block has travelled when it reaches the peg.

(3)

(Total 11 marks)

Q15.

A puck, of mass 0.2 kg, is placed on a slope inclined at 20° above the horizontal, as shown in the diagram.



The puck is hit so that initially it moves with a velocity of 4 m s^{-1} directly up the slope.

(a) A simple model assumes that the surface of the slope is smooth.

(i) Show that the acceleration of the puck up the slope is –3.35 m s^{-2}, correct to three significant figures.

(3)

(ii) Find the distance that the puck will travel before it comes to rest.

(3)

(iii) What will happen to the puck after it comes to rest?

Explain why.

(2)

(b) A revised model assumes that the surface is rough and that the coefficient of friction between the puck and the surface is 0.5.

(i) Show that the magnitude of the friction force acting on the puck during this motion is 0.921 N, correct to three significant figures.

(3)

(ii) Find the acceleration of the puck up the slope.

(3)

(iii) What will happen to the puck after it comes to rest in this case?

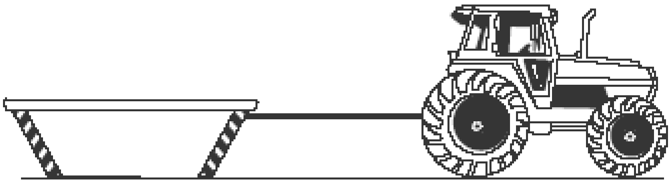
Explain why.

(2)

(Total 16 marks)

Q16.

A tractor, of mass 4000 kg, is used to pull a skip, of mass 1000 kg, over a rough horizontal surface. The tractor is connected to the skip by a rope, which remains taut and horizontal throughout the motion, as shown in the diagram.



Assume that only **two** horizontal forces act on the tractor. One is a driving force, which has magnitude P newtons and acts in the direction of motion. The other is the tension in the rope.

The coefficient of friction between the skip and the ground is 0.4.

The tractor and the skip accelerate at 0.8 m s^{-2}.

(a) Show that the magnitude of the friction force acting on the skip is 3920 N.

(2)

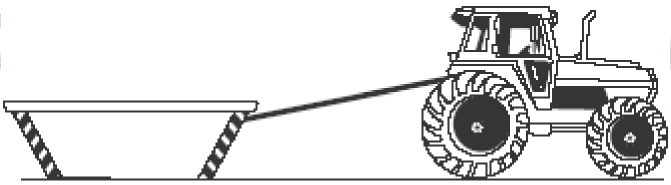
(b) Show that P = 7920.

(3)

(c) Find the tension in the rope.

(3)

(d) Suppose that, during the motion, the rope is **not** horizontal, but inclined at a small angle to the horizontal, with the higher end of the rope attached to the tractor, as shown in the diagram.



How would the magnitude of the friction force acting on the skip differ from that found in part (a)?

Explain why.

(2)

(Total 10 marks)

Q17.

Two particles, A and B, are connected by a light inextensible string, which passes over a smooth peg. Particle A is on a rough horizontal surface and has mass 3 kg. Particle B hangs freely, as shown in the diagram, and has mass 2 kg. The coefficient of friction between and the horizontal surface is μ.

The particles are released from rest and move with a constant acceleration of magnitude 0.9 m s^{-2}.

(a) Find the tension in the string.

(3)

(b) Draw and label a diagram to show the forces acting on particle A.

(1)

(c) Calculate the magnitude of the normal reaction force acting on A.

(1)

(d) Find the magnitude of the friction force that acts on A.

(2)

(e) Find μ.

(2)

(Total 9 marks)

Q18.

A block, of mass 5 kg, slides down a rough plane inclined at 40° to the horizontal. When modelling the motion of the block, assume that there is no air resistance acting on it.

(a) Draw and label a diagram to show the forces acting on the block.

(1)

(b) Show that the magnitude of the normal reaction force acting on the block is 37.5 N, correct to three significant figures.

(2)

(c) Given that the acceleration of the block is 0.8 m s^{-2}, find the coefficient of friction between the block and the plane.

(6)

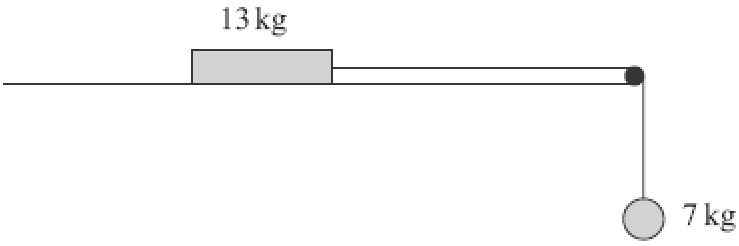
(d) In reality, air resistance does act on the block. State how this would change your value for the coefficient of friction and explain why.

(2)

(Total 11 marks)

Q19.

The diagram shows a block, of mass 13 kg, on a rough horizontal surface. It is attached by a string that passes over a smooth peg to a sphere of mass 7 kg, as shown in the diagram.



The system is released from rest, and after 4 seconds the block and the sphere both have speed 6 m s^{-1}, and the block has **not** reached the peg.

(a) State **two** assumptions that you should make about the string in order to model the motion of the sphere and the block.

(2)

(b) Show that the acceleration of the sphere is 1.5 m s^{-2}.

(2)

(c) Find the tension in the string.

(3)

(d) Find the coefficient of friction between the block and the surface.

(6)

(Total 13 marks)

Q20.

A box, of mass 3 kg, is placed on a slope inclined at an angle of 30° to the horizontal. The box slides down the slope. Assume that air resistance can be ignored.

(a) A simple model assumes that the slope is smooth.

(i) Draw a diagram to show the forces acting on the box.

(1)

(ii) Show that the acceleration of the box is 4.9 m s^{-2}.

(2)

(b) A revised model assumes that the slope is rough. The box slides down the slope from rest, travelling 5 metres in 2 seconds.

(i) Show that the acceleration of the box is 2.5 m s^{-2}.

(2)

(ii) Find the magnitude of the friction force acting on the box.

(3)

(iii) Find the coefficient of friction between the box and the slope.

(5)

(iv) In reality, air resistance affects the motion of the box. Explain how its acceleration would change if you took this into account.

(2)

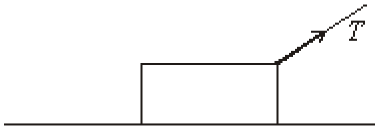
(Total 15 marks)

Q21.

A crate is being pulled at constant speed across rough horizontal ground by a rope.

The crate is of weight 100 newtons and the frictional force between the crate and the ground is of magnitude 30 newtons.

The tension in the rope is of magnitude T newtons.



(a) Draw and label a diagram to show all the forces acting on the crate.

(1)

(b) The coefficient of friction between the crate and the ground is 0.5. Show that the normal reaction force between the crate and the ground is 60 newtons.

(2)

(c) Explain why the horizontal component of the tension in the rope is 30 newtons.

(2)

(d) Find the value of T.

(4)

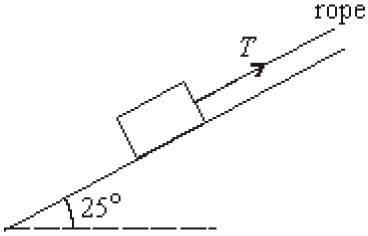
(e) Find the angle that the rope makes with the horizontal.

(3)

(Total 12 marks)

Q22.

A rough slope is inclined at an angle of 25° to the horizontal. A box of weight 80 newtons is on the slope. A rope is attached to the box and is parallel to the slope. The tension in the rope is of magnitude T newtons. The diagram shows the slope, the box and the rope.



(a) The box is held in equilibrium by the rope.

(i) Show that the normal reaction force between the box and the slope is 72.5 newtons, correct to three significant figures.

(3)

(ii) The coefficient of friction between the box and the slope is 0.32. Find the magnitude of the maximum value of the frictional force which can act on the box.

(2)

(iii) Find the least possible tension in the rope to prevent the box from moving down the slope.

(4)

(iv) Find the greatest possible tension in the rope.

(3)

(v) Show that the mass of the box is approximately 8.16 kg.

(1)

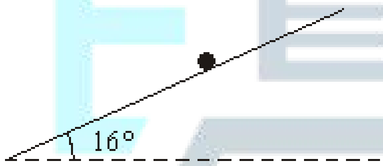
(b) The rope is now released and the box slides down the slope. Find the acceleration of the box.

(3)

(Total 16 marks)

Q23.

A stone rests in equilibrium on a rough plane inclined at an angle of 16° to the horizontal, as shown in the diagram. The mass of the stone is 0.5 kg.



(a) Draw a diagram to show the forces acting on the stone.

(1)

(b) Show that the magnitude of the frictional force acting on the stone is 1.35 newtons, correct to three significant figures.

(3)

(c) Find the magnitude of the normal reaction force between the stone and the plane.

(2)

(d) Hence find an inequality for the value of μ, the coefficient of friction between the stone and the plane.

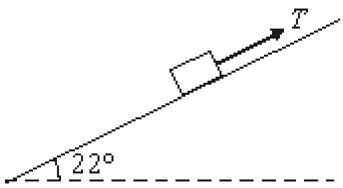
(2)

(Total 8 marks)

Q24.

A block is being pulled up a rough plane inclined at an angle of 22° to the horizontal by a rope parallel to the plane, as shown in the diagram.

The mass of the block is 0.7 kg, and the tension in the rope is T newtons.



(a) Draw a diagram to show the forces acting on the block.

(1)

(b) Show that the normal reaction force between the block and the plane has magnitude 6.36 newtons, correct to three significant figures.

(3)

(c) The coefficient of friction between the block and the plane is 0.25. Find the magnitude of the frictional force acting on the block during its motion.

(2)

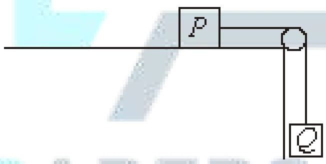
(d) The tension in the rope is 5.6 newtons. Find the acceleration of the block.

(4)

(Total 10 marks)

Q25.

A small block P is attached to another small block Q by a light inextensible string. The block P rests on a rough horizontal surface and the string hangs over a smooth peg so that Q hangs freely, as shown in the diagram.



The block P has mass 0.4 kg and the coefficient of friction between P and the surface is 0.5.

The block Q has mass 0.3 kg.

The system is released from rest and Q moves vertically downwards.

(a) (i) Draw a diagram to show the forces acting on P.

(1)

(ii) Show that the frictional force between P and the surface has magnitude 1.96 newtons.

(2)

(b) By forming an equation of motion for each block, show that the magnitude of the acceleration of each block is 1.4 m s^{-2}.

(5)

(c) Find the speed of the blocks after 3 seconds of motion.

(2)

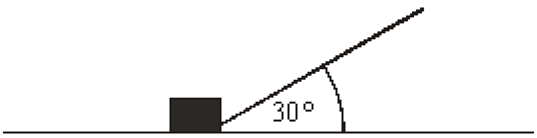
(d) After 3 seconds of motion, the string breaks. The blocks continue to move. Comment on how the speed of each block will change in the subsequent motion. For each block, give a reason for your answer.

(4)

(Total 14 marks)

Q26.

The diagram shows a rope that is attached to a box of mass 25 kg, which is being pulled along rough horizontal ground. The rope is at an angle of 30° to the ground. The tension in the rope is 40 N. The box accelerates at 0.1 m s^{-2}.



(a) Draw a diagram to show all of the forces acting on the box.

(1)

(b) Show that the magnitude of the friction force acting on the box is 32.1 N, correct to three significant figures.

(3)

(c) Show that the magnitude of the normal reaction force that the ground exerts on the box is 225 N.

(3)

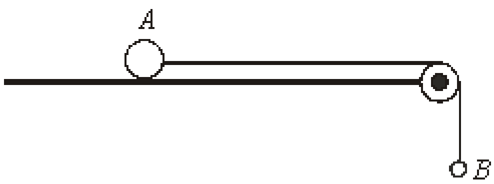
(d) Find the coefficient of friction between the box and the ground.

(2)

(Total 9 marks)

Q27.

The diagram shows two particles, A and B, that are connected by a taut light inextensible string. Particle A has mass 8 kg and is on a rough horizontal surface. Particle B has mass 7 kg and is attached to the other end of the string, as shown in the diagram. The string passes over a smooth light pulley. Particle A moves towards the pulley as particle B falls.



The coefficient of friction between A and the horizontal surface is 0.8.

(a) Find the magnitude of the friction force between A and the surface.

(2)

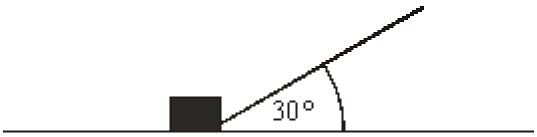
(b) By forming an equation of motion for each particle, show that the magnitude of the acceleration of each particle is 0.392 m s^{-2}.

(5)

(Total 7 marks)

Q28.

The diagram shows a rope that is attached to a box of mass 25 kg, which is being pulled along rough horizontal ground. The rope is at an angle of 30° to the ground. The tension in the rope is 40 N. The box accelerates at 0.1 m s^{-2}.



(a) Draw a diagram to show all of the forces acting on the box.

(1)

(b) Show that the magnitude of the friction force acting on the box is 32.1 N, correct to three significant figures.

(3)

(c) Show that the magnitude of the normal reaction force that the ground exerts on the box is 225 N.

(3)

(d) Find the coefficient of friction between the box and the ground.

(2)

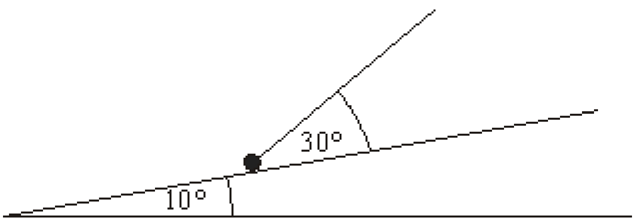
(e) State what would happen to the magnitude of the friction force if the angle between the rope and the horizontal were increased. Give a reason for your answer.

(2)

(Total 11 marks)

Q29.

A rough slope is inclined at an angle of 10° to the horizontal. A particle of mass 6 kg is on the slope. A string is attached to the particle and is at an angle of 30° to the slope. The tension in the string is 20 N. The diagram shows the slope, the particle and the string.



The particle moves up the slope with an acceleration of 0.4 m s^{-2}.

(a) Draw a diagram to show the forces acting on the particle.

(1)

(b) Show that the magnitude of the normal reaction force is 47.9 N, correct to three significant figures.

(4)

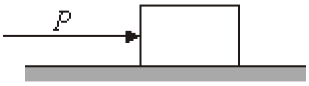
(c) Find the coefficient of friction between the particle and the slope.

(6)

(Total 11 marks)

Q30.

A small block, of mass 2 kg, is on a rough horizontal surface. The coefficient of friction between the block and the surface is $\frac{1}{7}$. A horizontal force of magnitude P newtons is applied to the block, as shown in the diagram.



(a) The magnitude of the frictional force that acts between the block and the surface is F newtons. Show that the maximum possible value of F is 2.8.

(2)

(b) The value of P increases gradually from 0 to 5.

(i) When P = 1, state the value of F.

(1)

(ii) When P = 5, the block moves with constant acceleration a m s^{-2}. Find the value of a.

(4)

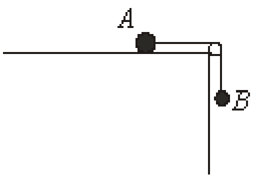
(Total 7 marks)

Q31.

A particle A of mass 0.2 kg is connected to a particle B of mass 0.3 kg by a light inextensible string.

The particle A is placed on a rough horizontal surface.

The string passes over a smooth fixed peg with B hanging freely, as shown in the diagram.



(a) The system is held in equilibrium by a horizontal force, P newtons, which is applied to the particle A. The coefficient of friction between A and the rough horizontal surface is 0.5.

(i) Draw a diagram showing the forces acting on the particle A.

(2)

(ii) Show that the magnitude of the maximum possible frictional force between A and the rough horizontal surface is 0.98 newtons.

(2)

(iii) Find the tension in the string.

(2)

(iv) Find the least value of P required to hold the system in equilibrium.

(2)

(b) The horizontal force P is removed and the particles start to move from rest.

(i) Find the magnitude of the acceleration of the particles during the subsequent motion.

(5)

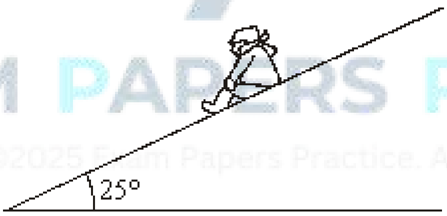
(ii) After falling 0.1 metres, the particle B hits the ground. Find the time between the system being released and B hitting the ground.

(2)

(Total 15 marks)

Q32.

A children's slide is straight and inclined at 25° to the horizontal, as shown in the diagram.



Matthew, of mass 35 kg, goes down the slide at constant speed.

(a) Draw a diagram to show the forces acting on Matthew.

(1)

(b) Find the magnitude of the normal reaction force between Matthew and the slide.

(3)

(c) Find the coefficient of friction between Matthew and the slide.

(4)

(Total 8 marks)

Q33.

A child kicks a small toy brick in a straight line across a horizontal floor. The brick initially moves at 3.5 m s^{-1} and comes to rest in a distance of 2.5 metres.

(a) Show that the magnitude of the retardation of the brick is 2.45 m s^{-2}.

(3)

(b) The mass of the brick is 0.2 kg.

(i) Find the magnitude of the frictional force acting on the brick.

(2)

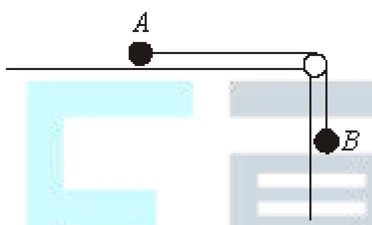
(ii) Find the coefficient of friction between the brick and the floor.

(2)

(Total 7 marks)

Q34.

Two particles, A and B, are connected by a light, inextensible string which passes over a smooth, fixed peg, as shown in the diagram.



The particle A is of mass 0.5 kg and the particle B is of mass 0.1 kg. The particle A is in contact with a rough horizontal surface.

(a) The system is at rest and the particle B hangs vertically.

(i) Show that the tension in the string is 0.98 newtons.

(1)

(ii) The particle A rests in limiting equilibrium. Show that the coefficient of friction between A and the surface is 0.2.

(3)

(iii) Draw a diagram to show the forces acting on the peg due to the string.

(1)

(iv) Find the magnitude of the resultant force on the peg due to the string.

(2)

(b) An additional mass of 0.1 kg is attached to the particle B and the system is released from rest. The particle A moves across the surface towards the peg and B moves vertically downwards. The particles move with acceleration a m s^{-2} and the tension in the string is T newtons.

(i) Show that a = 1.4.

(5)

(ii) Find the value of T.

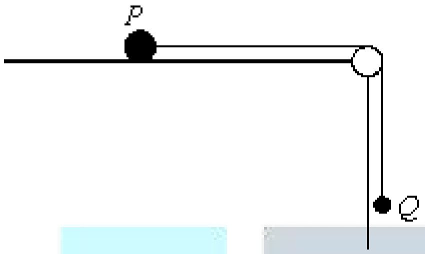
(1)

(iii) After falling a distance of 0.7 metres, B hits the ground. Find the time between the particles being released and B hitting the ground.

(2)
(Total 15 marks)

Q35.

Two particles, P and Q, are connected by a light, inextensible string which passes over a smooth, fixed peg, as shown in the diagram.



The particle P is of mass $2m$ and Q is of mass m. The particle P is in contact with a horizontal surface, which may be modelled as either rough or smooth.

(a) In the first case, the surface is modelled as rough and the particle Q hangs at rest. The coefficient of friction between P and the surface is μ.

(i) Find the tension in the string.

(1)

(ii) Find the range of possible values of μ.

(3)

(b) In the second case, the surface is modelled as smooth. The system is released from rest with the string taut and the particle P moves towards the peg.

Find the tension in the string.

(5)
(Total 9 marks)